Python Quick Reference

A Cheat Sheet for Core Concepts

Basics Data Types

Variables are dynamically typed. Indentation (4 spaces) is mandatory. Use ‘‘ for comments.

```
# Variable assignment
name = "Alice"  # String
age = 30        # Integer
height = 5.5    # Float
is_student = True # Boolean

# Print output with an f-string
print(f"{name} is {age} years old.")
```

Data Structures

Lists

Ordered, mutable sequences.

```
# List creation and indexing
nums = [10, 20, 30, 40, 50]
print(nums[0])      # -> 10
print(nums[-1])     # -> 50
print(nums[1:4])    # -> [20, 30, 40]

# List methods
nums.append(60)     # Add to end
nums.pop(2)         # Remove at index 2
```

Dictionaries

Unordered key-value pairs.

```
# Dictionary creation
person = {"name": "Bob", "age": 25}

# Accessing values
print(person["name"]) # -> Bob
print(person.get("city", "N/A"))

# Modifying
person["age"] = 26
person["city"] = "New York"
```

Common Operators

Category	Operators
Arithmetic	`+, -, *, /, %, **, //`
Comparison	`==, !=, <, >, <=, >=`
Logical	`and, or, not`
Assignment	`=, +=, -=, *=, /=`
Membership	`in, not in`

Control Flow

Conditional Statements

```
score = 85
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
else:
    print("Grade: C")
```

Loops

```
# For loop (iterating over a list)
fruits = ["apple", "banana", "cherry"]
for fruit in fruits:
    print(fruit)

# While loop
count = 0
while count < 3:
    print(f"Count: {count}")
    count += 1
```

Functions

Defining a Function

```
def greet(name, msg="Hello"):
    """Greets a person."""
    return f"{msg}, {name}!"

# Calling the function
print(greet("Eve")) # -> Hello, Eve!
```

Lambda Functions

```
# A lambda that adds 10 to a number
add_ten = lambda x: x + 10
print(add_ten(5)) # -> 15
```

File I/O

Use ‘with‘ for safe file handling.

```
# Writing to a file
with open("data.txt", "w") as f:
    f.write("Hello, World!\n")

# Reading from a file
with open("data.txt", "r") as f:
    content = f.read()
```

Pythonic Tips

- **List Comprehensions:** Concise list creation.
  `squares = [x**2 for x in range(5)]`
- **f-Strings:** Modern string formatting.
  `name="AI"; print(f"Hello, name!")`
- **Virtual Environments:** Isolate project dependencies.
  `python -m venv myenv`
- **Unpacking:** Assign iterable items to variables.
  `a, b, c = [1, 2, 3]`